Not Every Concept Is a Direction: Which Lexicalized Concepts Can Be Steered in LLMs? *

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Abstract

Steering vectors are widely used to control and interpret LLMs, yet benchmarks show that some concepts steer reliably while others barely respond — a variance that is reported but rarely explained. In this research we investigate which kinds of concepts are steerable in Llama-3.1-8B and Gemma-2-9B. We use the RFM method to extract steering vectors (or directions) for approximately 600 concepts drawn from 24 WordNet supersense categories, and validate which of them successfully steer the model. We then build a regression model explaining steerability from concept category, concreteness, polysemy, and word frequency, as well as the share of the top eigenvector during extraction (λ_1/trace). The results help distinguish concepts that behave as single directions — "primitive" for the model — from those that are composite or context-dependent, and offer practitioners an extraction-time indication of when single-vector steering can be trusted.

- **[NEW jul10, supersedes jul6 headline] Main run REDONE with the improved v4 extraction frame ("Imagine/assume: {c}."): 41.5% of the 600 concepts steer on Llama (249/600, V1$\cup$V6) and 37.3% on Gemma (224/600) — more than double the v2b rates — and the psycholinguistic laws now replicate across BOTH models with all four predictors significant: frequency +, concreteness –, age-of-acquisition –, polysemy –. Spectrum concentration (PR) predicts steerability within both models, at model-specific depths. Frequent, early-acquired, monosemous, abstract concepts steer best.**
- **[NEW jul6, superseded] v2b-instrument headline (kept for the instrument-comparison story): 119/600 (19.8%); zipf $+0.40^*$, conc -0.28^*; polysemy and AoA null under v2b.**
- Candidate sentence once results are in (pilot expectation): concreteness and polysemy — *not* frequency — predict steerability; the frequency null is itself a finding. (1 sentence)
- Candidate sentence (pilot mechanism): when steering fails, it fails informatively — single-vector steering delivers the concept's *category prototype* (a city steers to "a beloved place", not to itself). (1 sentence)
- Candidate sentence (causal probe): re-extracting with within-category contrasts tests whether such failures lie in the extraction design rather than the model. (1 sentence)
- Candidate sentence (replication): findings replicate across two model families (Llama, Gemma). (1 sentence)

*Working title. Alternative: "Concreteness and Polysemy Predict the Steerability of Lexicalized Concepts" (use if R2 confirms). **[NEW jul6] R2 did NOT confirm: polysemy is null and concreteness is negative; frequency is the positive predictor. Do not use the alternative title.**

1 Introduction (1.0p)

- Hook: steering/RePE papers throw heterogeneous concept lists at models; some concepts steer at 90%, others at chance. The variance is treated as noise; we argue it is signal. (1 para)
- The implicit assumption of top-1 eigenvector / PCA-1 / difference-of-means steering: the concept is representable as *one direction*, injectable with *one coefficient*, at whatever layer(s) the method uses. (half para)
- Existing evidence that not all features are one-dimensional (Engels et al.) — but no one has asked *which* concepts behave as single directions, or whether this aligns with human semantic structure. (half para)
- Our question: is steerability systematic and predictable from the linguistic properties of the word — and what does a steering failure actually consist of? (half para)
- Contributions (short lead-in + three numbered items, 2–3 sentences each): (1 para)
  - C1: a systematic steerability study — ∼600 lexicalized concepts across 24 WordNet supersenses, annotated with psycholinguistic norms, steered under a controlled two-configuration protocol; per-concept results and code released.
  - C2: a regression explaining steerability variance from concept category, concreteness, polysemy, and frequency, with extraction-internal spectral metrics (λ_1/trace, λ_1/λ_2) as candidate zero-cost predictors.
  - C3: a mechanistic account of what failure looks like, grounded in pilot diagnostics (parent-dominance of extracted directions; coherence $\neq$ identity) and tested with one causal probe (within-category contrast rescue).

2 Background and Related Work (1.0p)

- **Linear representation hypothesis and steering.** Park et al.; RepE (Zou et al.); CAA (Panickssery/Rimsky et al.); ActAdd (Turner et al.); ITI (Li et al.); the RFM/universal-steering benchmark we build on (Beaglehole et al. 2026) — kernel ridge + AGOP metric learning, top eigenvector as steering vector. *Ours:* we adopt this family's method (RFM) unchanged, but instead of proposing a steering technique we ask *which concepts* the technique works for and why. [**NEW jul10**] A second relation emerged: this literature's success rates are *instrument-relative* — changing one sentence of the extraction contrast doubles the steerable share (19.8%→41.5%) on the *same* method, model, and judge; benchmark comparisons that vary methods while hand-tuning per-class prompts inherit this confound (our per-class frame grid quantifies it). (1 para)
- **Multi-dimensional features.** Engels et al. (not all features are one-dimensional; circular features); feature splitting in SAEs; conceptors / subspace steering. *Ours:* same premise (one direction may not suffice), but we test it at the level of a stratified concept inventory and connect it to word properties rather than to individual features. [**NEW jul10**] Result: how concentrated a concept's discriminative spectrum is (participation ratio / λ_1-share) predicts steering success in both models — "effectively one-dimensional" is a measurable, graded property, and polysemy appears to act on steerability *through* it. (half para)
- [**NEW jul6**] **Hierarchical concept geometry — must cite.** Park, Choe, Jiang & Veitch (ICLR 2025, arXiv:2406.01506): ∼900 WordNet concepts on Llama-3 and Gemma; categorical concepts are simplices and hierarchically related concepts occupy *orthogonal* subspaces.

The representational-geometry twin of this study (same inventory source, same model families). *Ours:* they characterize how the hierarchy is *encoded*; we test what the hierarchy predicts about *intervention* — and our parent-dominance finding (extraction returns the shared category axis) is the steering-side counterpart of their parent/child orthogonal decomposition. **[NEW jul10]** R4 (averaging variant) has now run: subtracting the shared category axis does *not* rescue failed concepts (9 gained / 14 lost of 150) — consistent with their orthogonality result read causally: the child-specific residual exists but is rarely independently injectable; the shared axis also carries the steering *drive*. (half para)

- **Predicting steerability.** LAP / logit-lens accessibility (Billa 2026); "When is your LLM steerable?"; (un)reliability of steering vectors. *Ours:* these predict from engineering signals; we predict from the *linguistic structure* of the concept, and additionally explain what the failure mode consists of. **[NEW jul10]** Our lexical laws (frequency +, concreteness −, AoA −, polysemy −) replicate across two model families and two extraction instruments — portable, human-interpretable predictors that engineering-signal approaches cannot supply; conversely we contribute a caution their framing lacks: *geometric* predictors are only interpretable relative to the contrast design (our λ_1-share flipped from null to significant when the frame changed). (half para)
- **[NEW jul6] Add cites here:** (a) "Understanding (Un)Reliability of Steering Vectors in Language Models" (arXiv:2505.22637) — directional agreement between per-example activation differences and the steering vector predicts steerability; (b) its 2026 follow-up on *geometric predictors* of steering unreliability (arXiv:2602.17881) — closest neighbor to our R3; differentiate: they predict from activation geometry, we from the concept's linguistic structure plus extraction-time spectra, and we characterize the failure mode (parent delivery). (c) AxBench (arXiv:2501.17148) — the standard large-scale steering evaluation (prompting/finetuning baselines beat SAE steering); one line positioning our judge protocol and per-concept inventory relative to it. **[NEW jul10]** Positioning vs. AxBench sharpened: their Concept500 samples concepts *from the model* (SAE-latent labels — pre-selected to be represented, no lexical norms, no category stratification) and varies *methods*; we sample from the *lexicon* (WordNet-stratified, normed) and hold the method fixed — complementary designs, and ours is the one that can ask which kinds of concepts steer. Their trained ReFT-r1 also marks a boundary of our claims: "unsteerable" here means *extraction-based rank-1* steering fails; a supervised intervention might rescue some failures.
- **Causal identification of concept vectors.** Function vectors (Todd et al. 2024); linear relation embeddings (Hernandez et al. 2024); ROME (Meng et al.) — identification by *intervention*. *Ours:* complementary — we identify by *contrast design* and use one intervention-style probe (R4, now executed: no net rescue) to arbitrate when the contrast is at fault. **[NEW jul10]** The depth sweep adds a second point of contact with ROME's causal traces: entity content becomes injectable in a sharp mid-stack window (depths 14–17 of 32), mirroring where causal tracing localizes subject enrichment — read-side and write-side localization agree on *where*, while our sweep shows read-side statistics cannot substitute for write-side tests. (half para)
- **Frequency and linear representations.** Co-occurrence thresholds for linear representations; long-tail knowledge (Kandpal et al.). *Ours:* we test the frequency hypothesis directly. **[NEW jul10] Outcome reversed from the pilot expectation:** frequency is a robust *positive* predictor in both models — consistent with co-occurrence-threshold accounts (more exposure ⇒ cleaner linear structure) rather than with our pilot's "exposure alone buys nothing." The frequency effect thins when spectrum concentration enters the model, suggesting exposure works partly *through* representational consolidation. (half para)

- **LLM–human concept alignment and psycholinguistic norms.** Concreteness/AoA norms in NLP; Shani et al. 2025 ("From Tokens to Thoughts", arXiv:2505.17117): LLMs trade compression for meaning differently than humans, measured on *input-side* token embeddings vs. human categories. *Ours:* same alignment question, but measured mid-stream and *causally* — does human semantic structure predict what can be steered, not just how embeddings cluster. [**NEW jul10**] Answer: yes, and the psycholinguistic profile is coherent — words acquired *early* by humans and used frequently are the most steerable, over and above frequency (AoA $\beta = -0.33/-0.49$ across models); human acquisition order tracks which concepts LLMs consolidate into injectable directions. (half para)
- **Methodological caution on cosine.** You 2025 ("Semantics at an Angle", arXiv:2504.16318): cosine similarity can mislead in anisotropic learned spaces (a shared dominant component inflates all similarities). *Ours:* our geometry claims rest on *differential* cosines under identical measurement (within-class 0.84 vs. between-class 0.35 vs. moods 0.46), with the between-class value reported as the anisotropy control; mid-stream RMSNorm anchors the metric. (half para)
- Gap statement: no work connects concept-level semantic structure (concreteness, polysemy, taxonomic class) to steering outcome, nor explains what an unsteerable concept's failure consists of. [**NEW jul10**] This paper's answer, in one sentence each: (i) frequency, age-of-acquisition, polysemy, and concreteness jointly predict steerability, replicating across Llama and Gemma; (ii) failure is usually *informative* — delivery of the category/frame blend, a diffuse discriminative spectrum, or injection outside the concept's depth window — not silence. (half para)

3 Defining and Sampling Concepts (1.0p)

- **Operational definition.** Sidestep the philosophical debate (Murphy 2002; Margolis & Laurence 1999): a *lexicalized concept* = a meaning a speech community assigned a word to, operationalized as a WordNet synset; each word mapped to its dominant (first) noun sense. (half para)
- **Inventory.** All single-word, noun-dominant lemmas from WordNet; category = supersense (26 lexicographer files; Miller 1990; Fellbaum 1998; "supersense": Ciaramita & Johnson 2003). Exclusions: `noun.Tops`; POS-ambiguous lemmas; small cells sampled exhaustively and flagged. (half para)
- **Sample (≈600, flat).** 25/supersense across ∼24 usable supersenses; within each supersense, stratify over frequency so category and frequency are not confounded. Six supersenses are prespecified (before extraction) as the *featured* cells for within-category analyses: `location`, `person`, `feeling` (pilot continuity) and `artifact`, `cognition`, `food` (concreteness extremes). At n=25 every cell is adequately powered for class means and absorbs attrition (prompted-ceiling/judge exclusions). (1 para)
- **Predictors** (one table): Zipf frequency (wordfreq); concreteness (Brysbaert et al. 2014); age of acquisition (Kuperman et al. 2012); polysemy (#WordNet senses); taxonomic depth (covariate); supersense; log category size (covariate). (half para + table)
- **Contrast design.** Positives wrap a statement in a persona frame; the frame must be felicitous across *all* supersenses (e.g. "You are fascinated by {concept}"), else frame felicity confounds category. *Dual negatives*: (i) bare statement (benchmark-comparable); (ii) same frame with a *different* concept — free, since those prompts are other concepts' positives — which cancels the

frame axis, and with same-supersense negatives also the category parent. Rationale: a contrast in which frame, category, and instance always co-occur cannot statistically separate them (see Pilot). (1 para)

- **[NEW jul10] Main-run frame: v4 = "Imagine/assume: {c}." (constant prefix, concept unquoted; negatives bare).** Selected from a 7-frame grid on the pilot (Table 3, Appendix F): 77/90 V1∪V6 vs. 61 (orig class-specific) and 59 (v2b dual frame) — the only frame at ceiling on moods *and* far ahead on places. Ablations attribute the gain to *assume* (bare-noun reading ≈ role-assignment, a class-neutral persona prime) and show quoting the concept costs ∼7 points (mention-mode suppresses enactment). Trade-off, quantified in Appendix F: a constant prefix re-saturates extraction R^2 (the prefix is a free class separator), so the dual-frame diagnostics (informative val-r) are given up in exchange for steering power; the frame-leak caveat (R4) transfers to the imagine/assume register. **TODO: re-run the v_{spec} subtraction check on the v4 directions before camera-ready.**
- **Pilot/confirmatory split.** Pilot = 90 concepts (30 each: places, moods, personalities) from the released benchmark lists, used for all exploration; the WordNet sample is confirmatory, untouched during exploration. (half para)

4 Method (1.0p)

- **Direction extraction (RFM) — and why RFM.** Per layer: kernel ridge regression on concept-vs-negative prompts; AGOP $M = \mathbb{E}[\nabla f \nabla f^\top]$; steering vector = top eigenvector of M. Chosen among RepE methods because (a) it is the method of the benchmark we build on (comparability); (b) unlike difference-of-means or PCA-1 it returns the *full discriminative spectrum* with eigenvalues — the spectral metrics below come for free, and the method is self-diagnosing (the pilot's confound was discovered through them); (c) its steering performance matches or beats the simpler extractors on the released benchmark. (1 para)
- **Spectral metrics.** $\lambda_1/\mathrm{tr}(M)$ and λ_1/λ_2; per-concept score = band average over decoder depth 4–24 (per-layer profile retained). These measure the *coherence* of the extracted signature, not concept identity (see Pilot). (half para)
- **Fixed hyperparameters — a methodological requirement.** Kernel bandwidth inflates λ_1/trace mechanically (large bandwidth ⇒ near-linear fit ⇒ rank-1 AGOP); grid search over near-tied validation correlations injects artifact variance into the dependent measure. All extractions at bw=1, norm=True, reg=10^{-3}; vector and metrics read at a fixed (converged) iteration. (half para)
- **Two-config steering protocol.** Each concept steered under (i) *all layers*, benchmark defaults (unit vectors, released coefficient grid — comparability); (ii) *one mid layer* (≈40% depth; vector scaled by the concept's per-layer projection gap, cf. CAA / ITI; effective single-site dose ≈1× local residual-stream norm). Δ(all − single) measures how *distributed* the concept's steerable content is. 3 category-neutral prompts (listed verbatim); greedy decoding. (1 para)
- **Outcome.** Primary: fraction of (coef × prompt) cells judged as expressing the concept; explicitly *not* max-over-coefficients. Secondary: *granularity reached* (category / region / instance) — the binary criterion flattens near-misses (pilot: Edinburgh→"Scottish Highlands" scores 0). (1 para)

- **[NEW jul6] Graded judge (V8).** The 600-run generations show systematic sub-threshold movement the binary judge cannot credit (ragtime → lyre/melody imagery; candlepower → binoculars/glint): a graded rubric (1.0 named / 0.75 immediate field / 0.5 category level / 0.25 trace / 0 unrelated) re-judges the same generations and yields a continuous steerability outcome for the regression (thin-tail fix: only 119/600 binary positives).
- **Judge.** gpt-oss:20b (local, on the experiment GPU); symmetric judge question across supersenses. Validated against Claude Sonnet 5 on a 360-item set: 87–91% agreement, $\kappa = .72$–$.78$; a stricter candidate (Haiku 4.5) was rejected for one-directional divergence from the two-judge consensus. Confirmatory-study additions: re-validation on ∼200 WordNet-distribution items + human spot-check of disagreements (Appendix E). (half para)
- **Controls.** Prompted ceiling (can the model express the concept when explicitly asked?) — steerability conditional on ceiling; pipeline sanity-checked on known-good concepts per model. (half para)
- **Models.** Full protocol on Llama-3.1-8B-Instruct; headline analyses replicated on Gemma-2-9B-it. (half para)

5 Pilot Diagnostics and Protocol Selection (1.0p)

- **Hyper-parameter selection (summary).** Validation r saturates (∼0.97–0.99) across bandwidths, statement counts, and iterations ⇒ selection by validation is third-decimal noise; large bandwidth forces rank-1 AGOP by construction; converged readout cuts metric instability 3–4×. Decision rule: fixed modal setting, converged readout. Details in Appendix A. (half para)
- **Finding 1: parent-dominance.** Different same-category concepts' extracted top-1 directions are nearly identical (places: cosine 0.97 → 0.84 → 0.72 across depth; *all* top-3 components ≥0.90 shared). Single-layer dose-response saturates at the category *prototype* (every city → "quiet beach at sunset") with region-level near-misses (Edinburgh→Dublin pub); giving secondary components injection mass adds a generic roleplay axis, not instance content. Three legs: geometry, dose-response, component ablation. Interpretation: a contrast in which frame, parent, and instance always co-occur is statistically one axis — what steering delivers is mostly the category. (1 para + figure `parent_share_by_depth.png`)
- **Finding 2: coherence ≠ identity.** Places show the *highest* band λ_1/trace yet fail single-layer steering; moods (lower) succeed. Spectral metrics certify that a *coherent* (context-invariant, rank-1) direction was extracted — not that it is the judged content. Consequence: their regression coefficients are expected to be conditional on contrast type; tested under both contrasts. (1 para)
- **Finding 3: attribute vs. instance.** Attribute concepts (moods) steer from one mid layer; instance identities steer only under all-layer injection, reaching region granularity — consistent with multi-stage assembly (per-layer residue accumulation + late lexical tipping + autoregressive bootstrap once the name is emitted). Motivates the two-config protocol and Δ. (1 para)
- **Protocol justifications** (one line each; evidence in Appendices C–D): single mid layer at ≈40% depth — wide behavioral window there; 60% yields token-bias then repetition collapse. No top-k arms — the top-3 subspace under the released contrast is category-shared. Coefficient ≈1× local residual norm via per-layer magn scaling. Granularity as secondary outcome — the binary judge flattens near-misses. (1 para)

6 Results (1.5p)

- **R1: Steerability varies systematically.** Steering success varies widely across concepts and clusters by supersense; distribution figure per supersense, per config (Figure 3). (half para + figure)
- **[NEW jul10] R1 headline numbers (Llama, all-layer, v4 frame, judge = gpt-oss): 249/600 (41.5%) steer under V1∪V6** (V1 191, V6 195). Per-supersense spread 2–21 of 25: top `noun.feeling` (21), `noun.person` (17), `noun.animal` (16); bottom `noun.body` (2), `noun.shape` (4), `noun.quantity/noun.relation` (5). Relative to v2b (below), 22 of 24 supersenses improve and the ranking is broadly stable (feeling top, body bottom in both) — the instrument moves the *rate*, much less the *ordering*. `noun.body` is the one immovable cell (2/25 under both frames).
- **[NEW jul6, now the instrument-comparison row] R1 numbers under v2b (all-layer config).** 81/600 concepts steer under the subject-invite evaluation (V1), 95/600 under the neutral evaluation (V6), 119/600 (19.8%) under either; V1–V6 agreement $\phi = 0.59$ ($\kappa = 0.59$) — substantial but partly evaluation-relative, motivating the union outcome and the graded judge (V8). Per-supersense success spans 2–9 of 25 (no ceiling/floor): top `noun.feeling` (9), `noun.communication` (8), `noun.food` (7); bottom `noun.body` (2), `noun.plant` (4), `noun.artifact` (4). Figures: `plots/steered_by_supersense_600.png`, `plots/steered_boxplots_600.png`; analysis in `nb_analysis600.ipynb`.
- **R2: Psycholinguistic predictors.** Regression of steerability on concreteness, polysemy, AoA, Zipf, taxonomic depth (+ category size). Pilot expectation: concreteness and polysemy carry the signal; frequency ≈ 0 — exposure alone does not buy steerability. (1 para + table)
- **[NEW jul10] R2 outcome under v4 (Table 2): all four lexical predictors now significant, same signs in both models.** Llama: conc -0.20^{*}, zipf $+0.28^{*}$, AoA -0.33^{**}, polysemy -0.30^{**}; Gemma: conc -0.20^{*}, zipf $+0.23^{\dagger}$, AoA -0.49^{**}, polysemy -0.35^{**}. AoA and polysemy — null under v2b — surface once the instrument improves; the v2b signs (conc −, zipf +) hold. Nuance for honest reporting: polysemy and concreteness do *not* separate in raw univariate boxplots (`plots/covariate_boxplots_fv4.png`; MW $p>.6$) — their effects are conditional on the correlated covariates — whereas zipf, AoA, and PR separate univariately in both models ($p<.005$). Polysemy is largely absorbed by PR in the joint model: polysemous words appear to fail *via* diffuse spectra.
- **[NEW jul6] R2 outcome (Table 1): the pilot expectation is rejected on both counts.** Zipf frequency predicts steerability *positively* ($\beta = +0.40$, $p=.02$) and concreteness *negatively* ($\beta = -0.28$, $p=.017$, robust across specifications); polysemy and AoA are null. Frequent-but-abstract concepts steer best. Collinearity is benign among the lexical predictors (conc–zipf $r=.09$; all lexical VIF < 2.7).
- **R3: Spectral metrics as candidate predictors.** λ_1/trace and λ_1/λ_2 within the full regression; validity compared across the two contrast types (expected conditional — Pilot Finding 2); controlling for prompted ceiling and val-r (saturated ⇒ cannot explain variance). Baselines: frequency alone, category alone. (1 para)
- **[NEW jul10] R3 outcome under v4: ALL spectral metrics carry signal, and each alone beats the whole lexical block.** Llama (one-at-a-time): frac1 $+0.55^{**}$ (AIC 778.5), λ_1/λ_2 $+0.57^{**}$ (775.3), PR -0.49^{**} (786.5) vs. lexical block 799.0 (null 816.4); joint lex+PR: PR -0.62^{**}, best AIC 763.6. Under v2b, λ_1/trace was null — mechanism now understood

(Appendix F): with dual frames the top eigenvector sometimes latches onto the within-positive frame axis, corrupting spectral readouts; the constant v4 prefix removes that competition and the spectrum measures the concept. PR–frac1 collinearity is extreme under v4 ($r = -.98$): one-at-a-time is the only interpretable specification; never report the trio jointly.

- **[NEW jul6, v2b instrument] R3 outcome: the robust spectral predictor is the AGOP *participation ratio*, not λ_1/trace.** PR is negative both alone ($\beta = -0.27$, p=.023) and in the full model ($\beta = -0.72$, p=.007): concepts whose discriminative spectrum concentrates on fewer directions steer better. λ_1/trace has *no* independent effect (n.s. alone, sign-unstable in the trio; VIF = 12 within the spectral set) — quantitative confirmation of Pilot Finding 2 (coherence $\neq$ identity) at n=600: top-1 dominance per se does not buy steering. Report the one-spectral-at-a-time ladder alongside the full model (Table 1).
- **[NEW jul6] Category is mediated by measurable word properties.** Supersense dummies do not pay for themselves: AIC 626.7 (supersense-only) vs. 599.7 (null) vs. 576.7 (lexical+spectral, best); adding supersense to the best model *worsens* AIC (597.9). The category signal visible in R1 is carried by the continuous predictors — a stronger claim than "categories differ." **[NEW jul10] Softened under v4:** supersense-only now beats the null (AIC 784.1 vs. 816.4) — the wider per-class spread (2–21) carries real signal — but still loses to lexical+PR (763.6). Revised claim: category adds signal; continuous properties add more.
- **R4: The rescue probe (causal).** For instance-like concepts that fail: re-extract with within-category contrasts ("loves Edinburgh" vs. "loves {other city}"; frame and parent cancel) and steer at one mid layer. Success $\Rightarrow$ the failure was extraction addressing (contrast design), not model capability; failure $\Rightarrow$ instance content is not single-site injectable (assembly account). (1 para)
- **[NEW jul7] R4 status: the zero-reextraction averaging variant has been run (see end of this section); the full within-category re-extraction variant remains future work.**
- **R5: Distributedness.** Δ(all-layer − single-layer) per concept, reported descriptively by supersense and correlated with the R2 predictors. (half para)
- **[NEW jul10] R5 numbers under v4.** Single-layer depth-13 (magn dose): 90/600 (15.0%) vs. all-layer 249 (41.5%) — ratio 2.8× (v2b: 3.5×); the frame gain carries to single-site steering (34 $\rightarrow$ 90). 11 concepts steer *only* single-layer (over-steer rescues: cyanide, fearlessness, landmark, noggin, noonday, plasma, reciprocal, runt, segregation, sexism, trash — abstract/attribute skew, same diagnosis as the v2b run's 18). But see the depth-sweep subsubsection: depth 13 sits *below* the entity window, so the two-config Δ partly measures window placement, not distributedness alone — interpret Δ jointly with the sweep.
- **R6: Generality.** Llama vs. Gemma consistency: per-concept correlation of steerability and of the spectral metrics across models. (half para)
- **[NEW jul10] R6 outcome under v4: replication improves on every axis — cross-model convergence is partly instrument-dependent.** Rates: Gemma 224/600 (37.3%) vs. Llama 249 (41.5%); the v4 gain replicates (∼2× both models' v2b rates). Laws: all four lexical predictors significant with the same signs in Gemma (AoA -0.49^{**} strongest); PR replicates *at Gemma's informative depth* — null at Llama's 40%-depth readout, but -0.42^{**} (and frac1 $+0.39^{**}$) at 60–70% depth, surviving jointly with the lexical block (AIC 750.9 vs. 767.3): *same law, model-specific depth.* Items: per-concept $\phi = 0.43$ (154 shared) vs. 0.33 under v2b; supersense profile $r = 0.44$ vs. 0.12 n.s. Reading: part of the v2b item-divergence was instrument noise,

not model idiosyncrasy — a better instrument reveals *more* shared structure, though which concepts steer remains substantially model-relative (Gemma tops: event/phenomenon/group; body bottom in both). Gemma dose grid [14,17,20,23], shifted from the v2b run's [12,15,18,21] by dose-response analysis (Appendix D).

- **[NEW jul7, v2b instrument] R6 main-run outcome (Gemma-2-9B, 600 concepts): rates and laws replicate; items do not.** The overall steerable share replicates almost exactly (Gemma 113/600 = 18.8% vs. Llama 119/600 = 19.8%) and the regression signs replicate directionally (conc -0.19, p=.09; zipf $+0.31$, p=.06), but *which* concepts steer is largely model-specific: per-concept agreement $\phi = 0.33$ (53 shared, 66 Llama-only, 60 Gemma-only), the supersense profile does *not* replicate (r=0.12, n.s.; Gemma's top cells are cognition/process/phenomenon vs. Llama's feeling/communication/food), and cross-model spectral correlations are weak ($r \approx 0.13$–0.16). Framing: steerability is a *model-relative* property; what generalizes is the rate and the direction of the psycholinguistic laws, not the item list — consistent with directions being experiment- and model-relative (Discussion). Caveat: Gemma's coefficient window was calibrated on 6 pilot concepts; some item-level divergence may be dose noise.
- **[NEW jul6] R6 pilot evidence: behavior replicates, spectra do not.** On the 90-concept pilot (orig frames, V1) Gemma-2-9B matches Llama class-by-class (moods 30/30 both; personalities 23 vs. 24; places 0 vs. 2), with coefficients calibrated to Gemma's $\sim$50$\times$ larger residual norms (usable window 12–21). But the λ_1/trace class *ordering inverts* across models (Llama: places $>$ pers $>$ moods; Gemma: moods $\approx$ pers $>$ places) — spectral coherence is partly model-geometry, not concept-intrinsic, so cross-model spectral comparisons need within-model standardization. Gemma 600-concept replication in progress.
- **[NEW jul7] R4 outcome (averaging variant): no rescue effect.** We steered with $v_{\text{spec}}(c) = v(c) - \overline{v}_{\text{same supersense}}$ (leave-one-out, sign-aligned per layer, renormalized; no re-extraction) on six supersenses $\times$ 25 concepts (feeling, food, person, location, artifact, body), all-layer config, V1$\cup$V6: 25/150 steer vs. 30/150 baseline, with 9 failures flipped to success and 14 successes lost (McNemar $p \approx .4$ — no significant net effect in either direction; the steered set is otherwise stable, 16 retained vs. $\sim$5 expected under independence). Reading: *shared-component dominance is not the prevailing failure mode* — the raw directions are indeed $\sim$85% category-shared (mean $|\cos|$ to the supersense mean .81–.90), but removing that component rescues only $\sim$8% of failures, so most unsteerable concepts do not carry a masked concept-specific signal. Diagnostic side-yield (from inspection, not statistically tested): the subtraction visibly strips the v2b frame register from outputs, and several lost feeling-class successes (e.g. *exuberance*) had been credited on frame-injected affect rather than concept-specific content. Details and generations in Appendix C; the full within-category re-extraction variant remains future work.

7 Discussion (1.0p)

- What makes a concept steerable: attributes (context-general colorings) admit a single injectable direction; instance identities require assembled, multi-stage content — steering failure is usually *delivery of the parent category*, not silence. (1 para)
- What λ_1/trace is: an *intervention-match score* — how well the concept's natural activation signature fits the one move single-vector steering can make (one constant vector, all contexts). Mechanism for R2: concrete concepts express context-invariantly; abstract ones context-dependently. (half para)

	lexical only	lexical +PR	lexical +spectral
concreteness	$-0.20^{\dagger}$	-0.28^{*}	-0.45^{**}
Zipf frequency	$+0.40^{*}$	$+0.40^{*}$	$+0.33^{\dagger}$
AoA	-0.15	-0.08	-0.17
polysemy	-0.03	-0.02	$+0.08$
n tokens	$+0.08$	$+0.07$	$+0.03$
λ_1/trace			-0.18
λ_1/λ_2			$-0.44^{\dagger}$
participation ratio		-0.27^{*}	-0.72^{**}
AIC	591.3	588.0	576.7

Table 1: [**NEW jul6; v2b instrument — superseded by Table 2, kept for the instrument comparison**] Logistic regression of steerability (V1 ∪ V6) on z-scored predictors, n=600 (Llama-3.1-8B, all-layer, v2b frame). Reference AICs: null 599.7; supersense-only 626.7; supersense + all predictors 597.9. $^{\dagger}p<.1$, $^{*}p<.05$, $^{**}p<.01$. Spectral metrics are inter-correlated (PR–λ_1/trace $r=-.85$); the PR-only column is the conservative specification.

	Llama-3.1-8B		Gemma-2-9B	
	lexical	+PR	lexical	+PR$^{60\%}$
concreteness	-0.20^{*}	-0.43^{**}	-0.20^{*}	$-0.18^{\dagger}$
Zipf frequency	$+0.28^{*}$	$+0.23$	$+0.23^{\dagger}$	$+0.20$
AoA	-0.33^{**}	-0.28^{*}	-0.49^{**}	-0.47^{**}
polysemy	-0.30^{**}	-0.15	-0.35^{**}	-0.29^{*}
n tokens	$+0.04$	-0.05	$+0.06$	$+0.02$
participation ratio		-0.62^{**}		-0.39^{**}
AIC	799.0	763.6	767.3	750.9

Table 2: [**NEW jul10**] **Main regression (v4 instrument).** Logistic regression of steerability (V1 ∪ V6) on z-scored predictors, n=600 per model, all-layer config. PR read at each model's informative depth (Llama 40%, Gemma 60% — Gemma's spectral signal is null at the 40% convention and switches on at 60–70%). Reference AICs — Llama: null 816.4, supersense-only 784.1, one-spectral-alone 775–787; Gemma: null 794.8. Spectral collinearity is extreme under v4 (PR–λ_1/trace $r \leq -.94$ both models): spectral metrics are reported one-at-a-time only. $^{\dagger}p<.1$, $^{*}p<.05$, $^{**}p<.01$.

- [**NEW jul6**] **Revision required by R2/R3: the intervention-match story must move from λ_1/trace to the participation ratio, and the concreteness mechanism reverses.** Empirically it is spectrum *concentration* (low PR), not top-1 dominance, that predicts steering, and *abstract* (not concrete) concepts steer better — consistent with the pilot's parent-dominance reading: for concrete instance-like concepts the dominant direction is largely the shared category axis, so its coherence does not deliver the instance. (half para)
- [**NEW jul10**] **Second revision under v4: λ_1/trace is rehabilitated — its v2b null was an instrument artifact.** With the constant-prefix frame, top-1 dominance *does* predict steering ($+0.55^{**}$ alone) alongside PR; the v2b null arose because dual frames let the fascinated/expert register axis capture the top eigenvector for weak concepts (Appendix F), corrupting the spectral readout. General lesson for the field: *spectral diagnostics of extracted directions are only interpretable relative to the contrast design that produced them.* (half para)
- [**NEW jul10**] **Measured steerability is instrument-dependent — and so is cross-**

model generality. Changing one frame sentence doubles the steerable share in both models (19.8→41.5% Llama; 18.8→37.3% Gemma) and *raises* cross-model item agreement (ϕ 0.33→0.43) and category-profile agreement (r 0.12→0.44). Absolute rates from any single protocol should not be read as properties of the model alone; but the *laws* (frequency +, concreteness −, AoA −, polysemy −, spectrum concentration −) survive instrument change and model change, which is the paper's strongest generality claim. (half para)

- **[NEW jul10] Depth structure: concepts live in depth windows, ordered by abstractness.** Attributes are injectable from ∼30% depth onward with a broad plateau; entity identities switch on sharply at ∼45% (window 14–17 of 32) and fade late; personas need multi-site injection. One well-chosen layer nearly matches all-layer steering (71 vs. 77 of 90) — "all layers" mostly buys window coverage. Read-side extraction statistics do not locate the window (they miss the late decline entirely): injectability is a write property. (half para)
- Status of extracted vectors: the residual stream has no privileged basis, so extracted directions are *experiment-relative*; within-model relational claims are valid, coordinate-level and raw cross-model comparisons are not. Anchored spaces (embedding/unembedding, MLP neurons) are where mechanism pins a frame. (half para)
- Two routes to a vector's identity: *experiment design* (contrasts that vary exactly one thing) vs. *causal mediation* (function vectors, ROME). Pipeline: propose by contrast, accept by causal test — R4 instantiates it. (half para)
- Practical implications for RePE: report spectral concentration and contrast design alongside steering benchmarks; stratified concept inventories; granularity-graded judging over binary. (half para)

Future work: abstraction-depth as a second axis (0.5p)

- Idea: besides *how many* directions compose a concept, *where* (which layer) it first becomes steerable — hypothesis: hypernyms steer shallower than hyponyms (cf. pilot: moods at depth ∼9, places at ∼17+). (half para)
- Pilot lesson: this is a *write* property — extraction-geometry proxies fail the steering ground truth; measurement requires single-layer steering sweeps. Test on the WordNet hierarchy (taxonomic depth); independence from the compositional axis must be assessed *within* supersense. (half para)
- Related vignette: compositional concepts — is direction("strict father") ≈ α direction("strict") $+\beta$ direction("father")? (half para)

[NEW jul10, EXECUTED on the pilot] First-steerable-depth sweep

- **Design as run.** Pilot 90 (moods/places/personalities) × every single-layer depth 10–19, magn dose, v4 directions, V1∪V6 — ten eval passes, one local day (∼9 h). Figure 5. (half para)
- **Result 1: entity concepts have a sharp onset at depth 14 (of 32).** Places: 0–2/30 at depths 10–13, then **22/30 at depth 14**, window 14–17, fading by 19. This dissolves the earlier "places don't steer single-layer" conclusion: the old two-site protocol (13/19) *straddled the window.* (half para)

- **Result 2: attribute concepts steer shallow and everywhere.** Moods: 16/30 already at depth 10, plateau 25–30 from depth 13 through 19. The abstraction-depth hypothesis is confirmed ordinally: attributes become injectable ∼4 layers before instances. Personalities never find a single-layer home (peak 20, collapse to 1 by depth 19) — persona content appears genuinely multi-site. (half para)
- **Result 3: one well-chosen layer ≈ all layers.** Depth-15 alone: 71/90 vs. 77/90 all-layer — the all-layer advantage is largely *not knowing the window*, not distributed injection per se (except personalities). Consequence for R5: interpret Δ jointly with window placement. (half para)
- **What does NOT explain the window.** Per-layer extraction metrics (median val-R^2, frac1, PR, magn/hnorm) rise smoothly into depth 14 and stay flat through 19 (correlations with the sweep totals $|r| \approx 0.5$–0.6): they track the onset loosely but entirely miss the late decline — first-steerable-depth remains a *write* property not recoverable from read-side statistics (cross-layer orthogonality; Appendix Z2). (half para)
- **Confirmatory extension (future work):** same sweep on the six featured WordNet cells (150 concepts) with taxonomic depth as ground truth (hypernyms shallower than hyponyms?); independence from compositional rank assessed within supersense. (half para)

8 Limitations (0.5p; uncounted in ACL format)

- Single extraction method (RFM); λ_1/trace is AGOP-specific — PCA explained-variance as cross-check (appendix if time). (half para)
- Steering conclusions are relative to the additive intervention class; clamp/projection interventions may behave differently (targets saved, untested). (half para)
- LLM judge (validated but imperfect); English only; dominant-sense assumption; norms are human ratings. (half para)
- Correlational core with one causal probe (R4); extraction-design effects and model limits not fully separable outside that design. (half para)

A Appendix A: Extraction and metric setup

+ Validation r (probe accuracy) saturates (∼0.99) across bandwidths, statement counts, and RFM iterations ⇒ any selection *by* val-r is third-decimal noise; it is also a poor steerability predictor. Boxplots: `box_rsq.png`.
+ λ_1/trace at bandwidth L=1 discriminates across concepts; L=10 collapses it toward 1 (rank-1 AGOP by construction). `box_l1trace.png`, `box_l1l2.png`.
+ The released λ_1/trace (read at the best-r iteration) is unstable — it samples a value climbing 0.0002 → 0.17 → 0.46 across iterations at a noise-chosen point; reading at convergence cuts the statement-count swing 3–4×. Both vector and metrics read at a fixed (converged) iteration.
+ Saved per-layer extraction stats: top-3 eigenvalues (`evals`), natural concept magnitude (`magn` = raw projection gap), clamp targets (`tpos`, `tneg`), mean residual norm (`hnorm`) — in the rfmstats pickles.

? Per-concept "best layer" by argmax is unstable: val-r best is noise-driven; λ_1/trace and λ_1/λ_2 are bimodal with a degenerate spike at the first block (exclude early blocks). `besthist_*.png`.

? Hyper-parameter sweep factors (probe set of 24 concepts, one per supersense): bandwidth $\times$ normalization $\times$ contrastive construction (now incl. within-category negatives) $\times$ iteration read-out; criteria: stability, discrimination, semantic signal (Spearman vs. concreteness/polysemy), steering non-degradation. Artifacts: `hpsweep_phase2_*`.

B Appendix B: Predicting steerability (pilot 3 classes + 600-concept main run)

+ Best single predictor of steering success is λ_1/trace at a mid layer (depth 9, grouped CV-AUC $\approx$0.86); λ_1/λ_2 similar; val-r erratic (saturated). Logistic regressions in `nb_regxp1.ipynb`; `L9_metrics_by_steerability.png`.

– Multivariate regression on *all* layers' metrics is uninformative — severe multicollinearity (the residual stream is cumulative). Univariate per-layer CV-AUC is the clean read.

+ **Sign flips with depth**: λ_1/trace predicts steerability *positively* early (depth 9) but *negatively* late (depth 29) — late-layer concentration is lexical collapse, not concept concentration. `l1l2_depth29_dist.png`.

? Concept class is a confounder/mediator: most steerability variance is between-class; within-class signal is weak. Do not "control for class" if class is on the causal path (concreteness $\rightarrow$ signature coherence $\rightarrow$ steerability).

? **The inversion**: places have the *highest* band-averaged λ_1/trace yet fail single-layer steering, while moods (lowest) succeed — the metric certifies *coherence* of the extracted blend, not its *identity*. Predictor validity is contrast-dependent (main text, Pilot Finding 2).

? Participation ratio $\approx$2–3 for all concepts at all layers — the discriminative subspace at one layer is effectively rank 2–3, bounding what top-k can recover. `PR_depth9_hist.png`.

+ Error analysis at depth 9: misclassifications symmetric, in a narrow overlap band; characterizes the $\sim$0.86 ceiling.

+ **[NEW jul6] Main-run robustness: spectral collinearity.** The three spectral metrics are strongly inter-correlated at n=600 (PR–λ_1/trace $r=-.85$, λ_1/trace–λ_1/λ_2 r=.82; VIF up to 12), so joint coefficients are unstable. One-spectral-at-a-time ladder: PR alone $\beta=-0.27$ (p=.023, AIC 588.0); λ_1/λ_2 alone marginal (p=.086); λ_1/trace alone *null and sign-flipped* ($\beta=+0.02$, p=.89) — its large full-model coefficient is a collinearity artifact. PR is the reliable spectral predictor. `nb_analysis600.ipynb`.

+ **[NEW jul6] Main-run robustness: lexical predictors are clean.** Concreteness–Zipf r=.09; all lexical VIF < 2.7; the conc(−)/zipf(+) coefficients are stable across all specifications (with/without spectral terms, fixed vs. mixed effects).

+ **[NEW jul6] Main-run robustness: mixed-effects check.** Bayesian mixed logistic with a supersense random intercept reproduces the fixed-effects signs and magnitudes (PR $z=-6.6$, conc $z=-4.3$, zipf $z=+3.4$); supersense *fixed* dummies fail AIC against the continuous predictors (626.7 supersense-only vs. 576.7 lexical+spectral; adding dummies to the best model worsens it to 597.9).

+ **[NEW jul6] Outcome-robustness under the graded judge (V8).** Re-running the regressions with the graded score as outcome: concreteness ($-$) and spectral concentration (PR $-$, largest weight; spectral block ΔAIC ≈ 16) survive the outcome switch; the Zipf effect shrinks to marginal — part of the binary-judge frequency effect is plausibly *nameability* (frequent words surface verbatim and earn binary credit; V8 also credits semantic-field movement). Caveat: under V8, PR is significant only jointly with λ_1/trace, not solo — the conservative claim is "spectral shape adds predictive power, PR carrying most of it." Main text keeps the binary V1$\cup$V6 outcome (closest to the RFM benchmark's evaluation model); V8 is the secondary analysis. `nb_analysis600.ipynb`.

? **[NEW jul6] Note the pilot–main-run tension on λ_1/trace:** mid-layer λ_1/trace was the best pilot predictor (CV-AUC $\approx$0.86, 3 classes) but is null at n=600 under median-over-layers aggregation and all-layer steering. Candidate explanations: layer-specific vs. aggregated readout; 3-class between-class variance vs. 24-supersense within-class; contrast dependence (Pilot Finding 2). Worth one targeted analysis (per-layer readout on the 600) before the camera-ready. **[NEW jul10] RESOLVED:** the null was a v2b artifact — under dual frames the top eigenvector is captured by the within-positive frame axis for weak concepts (Appendix F, axis-alignment diagnostic); under the constant-prefix v4 frame λ_1/trace is significant alone ($+0.55^{**}$) with the pilot's sign.

+ Figures for this appendix: `L9_metrics_by_steerability` (Fig. 6); V8 validity scatter (Fig. 7).

+ **[NEW jul10] v4-instrument regressions (now the main numbers; Table 2 in main text).** Llama: all four lexical predictors significant (conc -0.20^{*}, zipf $+0.28^{*}$, AoA -0.33^{**}, polysemy -0.30^{**}); each spectral metric alone beats the entire lexical block (AICs 775–787 vs. 799; null 816); best model lex+PR (763.6, PR -0.62^{**}). Gemma replicates all lexical signs (AoA -0.49^{**} strongest) and PR at its own informative depth (60–70%: -0.42^{**} alone, -0.39^{**} jointly, AIC 750.9) while null at Llama's 40% convention — same law, model-specific depth. Univariate honesty note: polysemy and concreteness do *not* separate in raw boxplots (MW p>.6; effects are conditional); zipf/AoA/PR separate cleanly in both models (`plots/covariate_boxplots_fv4.png`). Spectral collinearity under v4 is extreme (PR–frac1 $r \leq -.94$): one-at-a-time only.

C Appendix C: Steering interventions (what moved the needle)

– Single-layer steering at depth 9: moods steer, experts middling, places $\approx$0 — places steer to a *generic* place attractor, not the named city (judge scores correct, not an artifact).

+ Two mid layers (9+17) was the best targeted multi-site result ($\sim$48/35 of 90); 9+26 worse (late layer contributes generic content); late-only steering fails outright.

+ **Cross-layer directions are near-orthogonal** (cosine $\approx$0.07): the concept is re-encoded per layer, so two layers are complementary slices while extra components at one layer are not.

– Single-site depth-19: *no behavioral window* — default $\rightarrow$ lexical word-lists (bitter $\rightarrow$ "bitter, cold, dry, sarcastic") $\rightarrow$ repetition collapse within 0.5–2$\times$ local stream norm; moods 10/15 (partly lexical credit), places/personalities $\approx$0. Late injection biases output *tokens*, not behavior.

+ Single-site depth-13: *wide* behavioral window (embodied persona, clean saturating dose-response, no collapse): moods 15/15; places 0/15, saturating at the category prototype ("quiet beach at sunset" for every city) with region-level near-misses (Edinburgh$\rightarrow$Dublin pub; genomicist$\rightarrow$neuroscientist). Coefficient is not the limiting factor; *vector content* is.

– Top-k at one layer, k=3: with eigenvalue weights ≈ k=1; with equal weights *worse* (moods 15→10) — components 2–3 are a generic roleplay/self-narration axis ("my name is Luna"), 0.92/0.90 shared across cities. **Parent-dominance**: all top-3 components ≥0.90 shared across same-category concepts at every depth (places cosine 0.97 → 0.84 → 0.72; moods individuate to 0.46) — the contrast, in which frame, parent, and instance always co-occur, is statistically one axis. `plots/parent_share_by_depth.png`.

+ All-layer steering (benchmark defaults): 26/45 v1; places reach *region* granularity (Edinburgh→Scottish Highlands with Scots dialect; Austin named exactly) — per-layer instance-residue accumulation + late lexical tipping + autoregressive bootstrap (once the name is emitted, context sustains it).

? Sample note: the depth-13/19, k=3, and all-layer bullets above report the interim n=15/class shakedown; all four arms to be re-run on the standard 90-concept pilot (extraction in `direction3e`; generation+judging only). Qualitative patterns expected to hold.

? **Planned probe — within-category contrast extraction** (= R4, full variant, not yet run): positives "loves Edinburgh", negatives "loves {other city}" (same frame and statements both sides) — frame and parent cancel; the only label-varying signal *is* the city. Steer single-layer (depth 13, magn-calibrated) on ∼5 cities. Cities named ⇒ failure was extraction *addressing*; still generic ⇒ instance content is not single-site injectable (assembly account).

– **[NEW jul7] Zero-reextraction averaging variant of R4: run at n=150; no rescue effect.** Method: per layer, unit-normalize the K=1 direction, sign-align within supersense (saved eigenvector signs are arbitrary — 4–28% of within-category pairs have negative cosine at mid layers; alignment via the top eigenvector of the 25×25 Gram matrix), subtract the leave-one-out category mean in the concept's own orientation, renormalize (dose unchanged). Six supersenses × 25 (feeling, food, person, location, artifact, body), all-layer default config, V1∪V6, tag `Lall_K1ev_def_fv2b_r4spec`, builder `build_r4spec_directions.py`. Result: 25/150 vs. 30/150 baseline; +9 rescued / −14 lost / 16 retained; McNemar $p \approx .4$; per-supersense cell movements (e.g. feeling 9→6, body 2→3) are all within noise at n=25/cell and are not interpreted. The raw directions are ∼85% category-shared (mean | cos | to the supersense mean .81–.90 across layers), yet removing that component flips only 9/120 failures — masked concept-specific signal exists (*beard*: baseline collapses into excited stammering, v_{spec} at c=0.8 yields on-topic beard-styles content; similarly *hay*, *fabric*, *residence*) but is the exception, not the prevailing failure mode.

+ **[NEW jul10] Depth sweep (pilot 90, v4 directions, single-layer magn, depths 10–19).** Totals 31/33/23/38/64/71/54/61/42/34 (d10→d19; V1∪V6). Places: 0–2 through d13, **22/30 at d14**, window 14–17 — the earlier two-site protocol (13/19) straddled it, producing the false "places don't steer single-layer." Moods: onset ≤10, plateau 25–30 from d13 through d19. Personalities: no single-layer home (peak 20, collapse to 1 at d19). Peak single-site d15 = 71/90 vs. all-layer 77. Per-layer read metrics (val-R^2, frac1, PR, magn/hnorm) rise into the window but stay flat through the late decline ($|r|$≈.5–.6 with totals): the window is not recoverable read-side. `plots/depth_sweep_fv4.png`, RUNLOG 021.

+ **[NEW jul10] Single-layer arm of the v4 600-run** (depth 13, magn): 90/600 (15.0%) vs. all-layer 249; ratio 2.8× (v2b: 3.5×); 11 single-layer-only rescues (abstract/attribute skew). NB depth 13 sits below the entity window found by the sweep — the confirmatory single-site arm should move to depth 15. RUNLOG 020.

? **[NEW jul7] Side-yield (inspection-level): v_{spec} doubles as a frame-leak control.** Baseline all-layer outputs carry the v2b register ("*excitedly* …fascinated …the wonders of") at all coefficients; v_{spec} outputs do not — confirming the frame component lives in the category mean. Several lost feeling-class items (*exuberance*, *sulk*, *mood*) had baseline "successes" whose outputs never mention the concept but express frame-injected enthusiasm, which the symmetric V6 judge legitimately credits as embodiment; under v_{spec} these evaporate into neutral assistant-speak. Observation from reading generations, not a tested effect — a targeted design (affect-blinded judge, larger feeling sample) would be needed to quantify how much of the feeling supersense's top R1 rank is frame-carried.

D Appendix D: Coefficient calibration (systematic magnitude)

– Fixed/hand-swept coefficients are brittle: the working value shifts $\sim 10\times$ with depth and with the number of steered layers.

\+ `magn` (natural projection gap, saved per layer) grows ∼exponentially with depth; scaling each layer's vector by `magn` makes the coefficient a dimensionless multiplier (cf. CAA mean-difference magnitude; ITI σ-scaling). Fig. 8.

\+ $\sqrt{N}$ count-normalization (cross-layer directions ≈orthogonal ⇒ quadrature) makes the multiplier roughly transfer across layer counts; usable band centers near ∼1.

\+ Single-site calibration: effective single-layer steering needs ≈1.0× the local residual norm (`hnorm`); `magn`/`hnorm` ≈ 0.4–0.5 mid-stream ⇒ multiplier ≈2–3. Empirically `magn` ≈ norm-relative scaling (r=0.90 with `hnorm`), so the two schemes are redundant.

? Clamp / project-to-target (set the projection to the concept's natural positive-class value; `tpos`/`tneg` saved) is the one structurally different alternative — bounded, token-dependent, self-allocating across layers; untested.

? Process lesson: three config-state incidents (stale judge CSVs, judge-cache pickles, stale coefficient list) share one root cause — hand-set knobs without provenance. Fix: auto-derive coefficient grids from saved per-layer stats; encode a run tag in artifact filenames.

\+ **[NEW jul10] Gemma dose-grid shift by dose-response audit.** Reading which coefficient produced each v2b-600 success: successes concentrate monotonically toward the top of the [12,15,18,21] grid (concept surfaces at 12 for only 1–3 concepts; *uniquely* at 21 for 12–13), and degeneration (repetition >25%) only begins at 21 (2.5–4% of outputs) — the grid truncated the dose-response on its rising side. v4 600-run grid shifted to [14,17,20,23]; best-over-coefficients scoring makes the added top dose risk-free.

\+ **[NEW jul6] Gemma-2-9B cross-model calibration.** Gemma-2's residual norms are ∼50× Llama's (median `hnorm` 484 vs. 9.4; $\sqrt{d}$ embedding scaling), yet the usable all-layer coefficient window is 12–21, *not* 50× Llama's 0.5–0.8: injection compounds over 41 layers and generation degenerates into repetition above ∼25. Grid [12, 15, 18, 21] reproduces the Llama pilot class-by-class (moods 30/30, personalities 23/30, places 0/30) — evidence that norm-relative reasoning brackets but does not pin the coefficient; a short empirical sweep remains necessary per model family.

E Appendix E: Judge selection and validation

+ Judge shootout on a fixed 360-item set (the pilot all-layer run's generations, v1+v4): local gpt-oss:20b 26/45 (v1), pod gpt-oss:20b 27/45, Claude Sonnet 5 26/45 — with *identical* per-class breakdowns for local-oss and Sonnet (10/15/1). Judge is reproducible across machines and agrees with an independent frontier model.

– Claude Haiku 4.5 is a systematically *strict outlier* (13/45 v1; moods 15→5): of 49 disagreements with Sonnet, 48 are one-directional (Haiku under-credits). Cause: it enforces fluency/coherence as a hard criterion, failing fragmented-but-clearly-on-concept outputs. Rejected — and a useful demonstration that judge choice can move absolute rates ∼2× while the class structure survives.

+ Primary-judge validation numbers (gpt-oss vs. Sonnet, n=360): agreement 86.7%/90.6% (v1/v4), Cohen's $\kappa = 0.72/0.78$ — substantial agreement. Artifacts: `oss_judgments*.json`, `sonnet_judgments*.json`, `haiku_judgments*.json` (per-concept, per-coefficient scores).

? Planned for the confirmatory study: (a) re-validate on ∼200 items drawn from the *WordNet* generations (the banked set covers only the 3 pilot classes; the 24-supersense distribution is broader); (b) human spot-check of the gpt-oss/Sonnet *disagreements* (∼25 items) to anchor the consensus against human judgment.

+ Practical: gpt-oss:20b runs on the experiment GPU (ollama, 100% GPU, ∼1.5 s/judgment) — judging the full study costs GPU-hours, not API credits.

F Appendix F [NEW jul6]: Frame and evaluation-prompt selection (pilot, N=90)

+ **Frame grid** (pers/moods/places on V1): `orig` class-specific 24/30/2 (56); `v2` "fascinated by / deeply preoccupied with" 24/22/10 (56); `v2b` "fascinated by / an expert on" 22/17/14 (53); `v3` "thinking about / on your mind" 18/16/7 (41). The universal frame is not a free lunch: it is required for the 24-supersense inventory (no class template exists for arbitrary nouns) and it *sets the construct*.

– **Frame-vocabulary leak (v2).** The dual frame cancels only what *differs* between its two variants; their shared semantics (fascinated ∩ preoccupied = intense absorption) stays in the class mean and leaks into every direction — steered moods embody *distraction* ("can't seem to focus…taps forehead"), costing moods 30→22 while lifting places 2→10 (the leak replaced the old persona-parent).

– **Weakening the verbs does not fix it (v3).** Semantically light frames produce *fainter* directions, not cleaner ones: under-steering everywhere (41/90; bitter → "a bit distracted"), worst of all frames. Intensity is the signal.

+ **Register split works partially (v2b).** Pairing an affective with an epistemic verb shrinks the shared core to "strong engagement": places best-ever (14 V1 / 13 V6) and balanced scores across classes under V6 (13/14/13 vs. v2's 9/24/3), at a cost to moods (22→17). *The frame acts as a register selector; no single frame is optimal for both attribute-like and entity-like concepts.* v2b chosen for the main run: the inventory is dominated by entity-like nouns, and a *balanced* instrument avoids pre-determining which supersenses can score (only `noun.feeling` pays the affect cost, quantified here).

+ **[NEW jul9] The v4 family: a single constant hypothetical prefix beats everything.** `v4` = "Imagine/assume: {c}." (concept unquoted, negatives bare) reaches **77/90** on V1∪V6 — the only frame at ceiling on moods (29–30) *and* far ahead on places (16 V1 / 22 V6, vs. 2–4 under `orig`) — while personalities are frame-insensitive (21–26 everywhere). Ablations (Table 3): quoting the concept costs ∼7 points, concentrated in the embodiment classes (`v4c` 66 vs. `v4b` 59; moods 28→19 on V1) — quotes mark metalinguistic *mention* and suppress enactment; dropping "/assume" costs ∼11 points, almost all in places (`v4` 77 vs. `v4c` 66); and the slash construction has no magic of its own — "Imagine/consider" is the *worst* of the family (`v4d` 56). Reading: with a bare noun, "assume: X" resolves as role-assignment ("assume the role of X") — a class-neutral persona prime, i.e. the benchmark's "Take on the role of" packaged inside a universal hypothetical; "consider" substitutes analysis and drags moods down (21/17). This revises the v3 lesson: low-affect frames *can* work when the prefix is *constant* (its content sits in the class mean and supplies drive) rather than dual (its content cancels). Caveats: (a) all constant-prefix frames re-saturate extraction R^2 (.985–.998), so val-r loses its predictive value and the top-eigenvector/frame-axis competition of dual frames cannot occur — but the frame-leak caveat returns in a new (role-adoption) register, so the v_{spec} subtraction check should be re-run before trusting v4 rates; (b) single runs at n=30/class (±3–5 judge noise): 59-vs-56 are ties, 77-vs-66 is structure. **[NEW jul10] v4 promoted to the main-run instrument**: 600-concept runs redone on both models (Llama 249/600, Gemma 224/600; RUNLOG 019/022); v2b results retained throughout as the instrument-comparison arm.

+ **[NEW jul9] Why dual frames depress val-r — the axis-alignment diagnostic.** Same 90 concepts, same code: orig frames give median per-concept val-R^2 .997, v2b .716 (paired Δ=.34); an old-vs-current-code rerun is exactly bit-equivalent (cos 1.0000, $\Delta \mathrm{R}^2$ −0.0000), exonerating the pipeline. Recomputing activations for the 4 lowest- and 4 highest-R^2 WordNet concepts: for high-R^2 concepts the saved top-1 direction aligns with the label axis ($|\cos|$ .65–.87) and not the frame axis (.01–.25); for low-R^2 concepts it aligns with the fascinated-vs-expert *frame* axis (.81–.93) and not the label axis (.04–.25) — the deliberately-inserted within-positive frame variance wins the top eigenvalue slot when the concept is weak, so K=1 injects a register axis. Notably the raw class-mean difference is healthy for *all* eight (norms 5.5–7.3, consistency .90–.94): the failure is eigenvalue *ranking*, not absent signal — and for the entangled cases (princess: $|\cos|$(concept, frame axis) = .37 vs. .04–.06 for clean concepts) the concept's own effect is register-colored, exactly "fascinated by a princess ≠ expert on princesses." Fix candidates: select the saved component by label correlation instead of eigenvalue; or constant-prefix frames (v4), which remove the competition.

+ **Evaluation versions.** V1 = category-inviting question ("What is your favorite subject?") — collapses the response space so a weak direction only tips a forced choice; V6 = neutral open field ("What would you like to talk about today?") + symmetric judge crediting aboutness *or* embodiment — only strong topic-steering surfaces. V1–V6 agreement on the 600-run: $\phi = 0.59$, $\kappa = 0.59$, Jaccard 0.48 — measured steerability is partly evaluation-relative; the main outcome is their union (cued ∨ uncued).

– **Zero-shot judge (V6Z) rejected.** Removing the judge's three in-context examples lowers or ties every class (40/90 → 34/90 on v2b) — the examples calibrate rather than anchor; the worry that a places-negative example suppresses places scores is empirically dismissed (places 13 under both).

? **Graded judge (V8).** Binary judging flattens systematic sub-threshold movement visible in the 600-run generations (ragtime → lyre/melody; candlepower → binoculars/glint): V8 re-judges the

frame	class	V1	V4	V6	V1∪V6	med R^2
`orig` (class-specific)	pers.	24	4	17	26	.996
	moods	30	27	27	30	.996
	places	2	14	4	5	.998
`v2` "fascinated / preoccupied", quoted	pers.	24	3	9	24	.804
	moods	22	16	24	26	.740
	places	10	4	3	11	.807
`v2b` "fascinated / expert on", quoted	pers.	22	–	13	23	.715
	moods	17	–	14	19	.785
	places	14	–	13	17	.531
`v3` "thinking about / on your mind", quoted	pers.	18	0	8	20	–
	moods	16	13	9	17	–
	places	7	9	5	8	–
`v4` "Imagine/assume: {c}."	pers.	23	9	9	23	.997
	moods	29	28	27	30	.995
	places	16	17	22	24	.992
`v4b` "Imagine: '{c}'." (quoted)	pers.	20	2	10	23	.997
	moods	19	20	14	22	.986
	places	10	13	12	14	.990
`v4c` "Imagine: {c}."	pers.	23	4	10	25	.997
	moods	28	27	24	28	.993
	places	8	12	12	13	.994
`v4d` "Imagine/consider: {c}."	pers.	21	3	5	21	.996
	moods	21	23	17	22	.985
	places	10	10	13	13	.991

Table 3: **[NEW jul9] Extraction-frame grid** (pilot N=90: 30 personalities / 30 moods / 30 places; Llama-3.1-8B, all-layer K=1, default coefficients, gpt-oss judge). Cells = concepts steered out of 30 under each evaluation version; V1∪V6 = union outcome; med R^2 = class median of per-concept median validation R^2 across layers. Constant-prefix frames (`orig`, v4 family) saturate R^2 by construction (the prefix is a free class separator); dual frames (`v2`, `v2b`) make R^2 concept-informative; `v3` direction files were not retained, so its R^2 cells are unavailable. V1∪V6 totals: `orig` 61, `v2` 61, `v2b` 59, `v3` 45, `v4` **77**, `v4b` 59, `v4c` 66, `v4d` 56.

same generations on a 0/0.25/0.5/0.75/1 semantic-proximity rubric, yielding a continuous outcome (thin-tail fix: 119/600 binary positives). Validation vs. V6 pending. `evaluation_prompts/wordnet_eval_`

? Judge-model selection is Appendix E; the full frame/eval run history with per-run tags is `RUNLOG.md` (runs 002–009, 015–018). Figures: val-R^2 by layer under v2b (Fig. 9); val-R^2 by steering outcome (Fig. 10).

G Appendix Z1 (parked): Localizing the concept — "where is Paris?" (diagnostics)

+ **ROME causal trace** ("Paris is the capital of"→France): the fact lives at the *subject* token in early–mid layers (∼0–16) and is relayed to the final token only late (∼14–31). `ROME-PARIS.png`; `diag_paris_rome.ipynb`; `diag_paris_rome_ha.ipynb`.
+ This explains the RFM mismatch: RFM reads the *last* token, where subject-specific content arrives only late ⇒ early extraction captures an *abstraction*, late the *precise* instance.

+ **SAE features** confirm the model holds Paris: clean Paris / France / city features at the steered layers (Goodfire l19; Llama-Scope L9/17/19, Neuronpedia labels). `diag_paris_sae.ipynb`, `diag_paris_sae_2.ipynb`. The steering failure is extraction/geometry, not absence.

? **Tegmark spatial probe** (lat/lon linear probe): coordinates linearly decodable but underpowered at N=90 cities (median ∼2200 km). `diag_paris_tegmark.ipynb`.

+ **Where "late" begins**: logit-lens KL-to-output gives stage boundaries to avoid the residual-sharpening tail. `diag_stages_late.ipynb`.

H Appendix Z2 (parked): Conceptual threads

? **Abstract-early / precise-late hierarchy**: the extraction layer sets the abstraction level; the precise concept lives late (weak steering) while the strong injection point is mid (concept abstract) — no single layer gives both.

? **Two candidate axes** (now Discussion future work): (i) abstraction-depth — first layer that can *steer* the concept; (ii) compositional basis size (λ_1/trace). Between-class confounded (moods both more composite *and* earlier-steering than places); independence is a within-supersense question.

– Extraction geometry cannot proxy first-steerable-depth: rotation and alignment-to-deep proxies are concept-invariant (∼17–20) and fail the steering ground truth (cross-layer orthogonality, $\cos \approx 0.05$). A write property; measure by steering. `nb_axes.ipynb`; `plots/axes_independence.png`.

? **Superposition vs. composition**: a "combinatorial basis × layer-specific sets" account is not ruled out; the SAE feature-count argument is the crux — open, not settled.

? **Stages of inference / lens** framing (Lad–Gurnee–Tegmark) as the principled naming of early/mid/late bands.

? **Not-all-linear** (Engels et al., arXiv:2405.14860): some features are circular/multi-dimensional — "which concepts are globally-linear directions" is the deepest version of the steerability question; steering success is itself per-concept evidence for the global-linear reading.

Figure 1: **[Figure 1 candidate — data to be regenerated for the v4 run.]** (A) Same protocol, opposite outcomes: dose-response for a steerable concept (*frog*) vs. an unsteerable one (*fairway*). (B) Share of concepts steered per WordNet supersense (V1∪V6). Options 2 (cross-model scatter) and 3 (pipeline schematic) in `plots/fig1_opt{2,3}.png`.

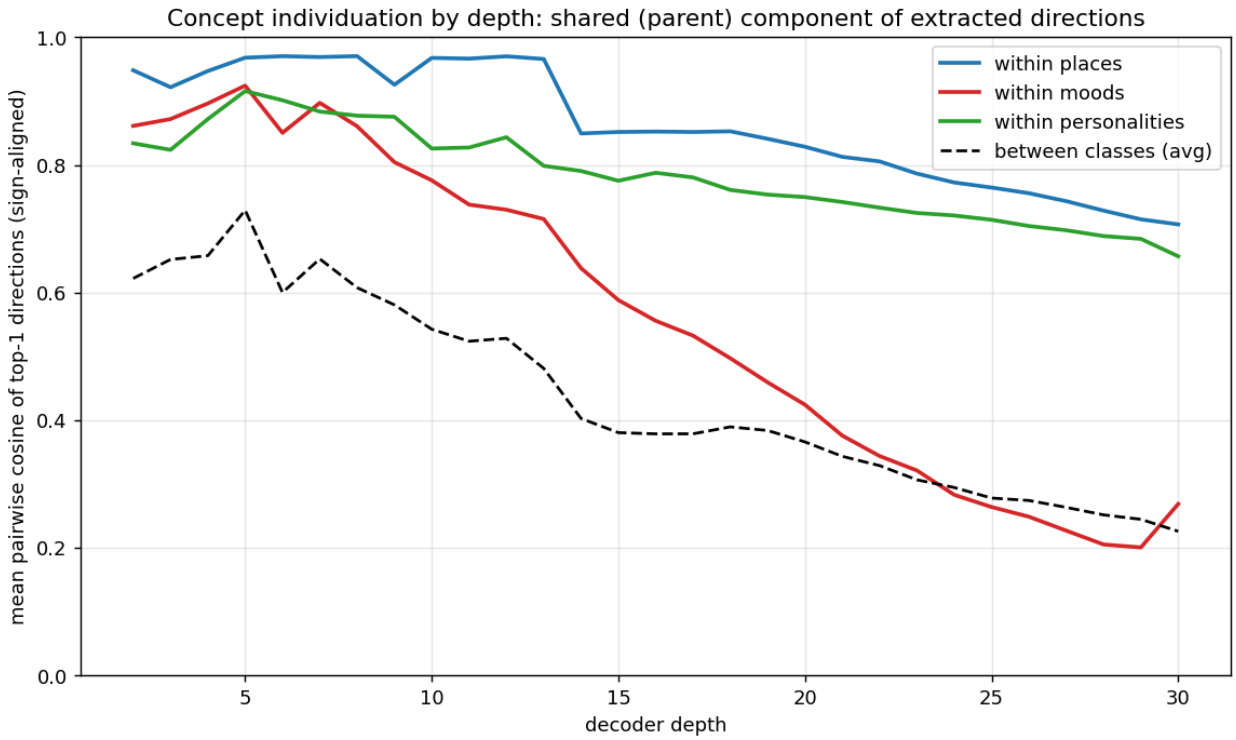

Figure 2: **Pilot Finding 1 (parent-dominance).** Cosine similarity of same-category concepts' extracted top-1 directions across depth: the released contrast yields largely *shared* (category) directions for places, while moods individuate.

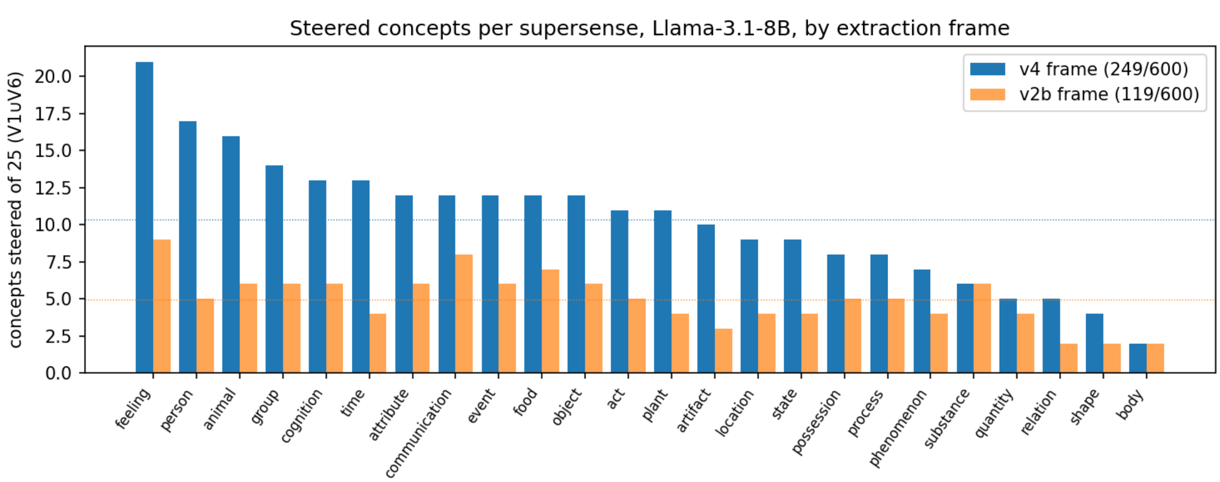

Figure 3: **[NEW jul10] R1: steered concepts per supersense** (of 25; V1∪V6; Llama-3.1-8B, all-layer config), under the main v4 instrument and the v2b instrument-comparison arm. The frame moves the rate (∼2×), far less the ordering (feeling top, body bottom under both).

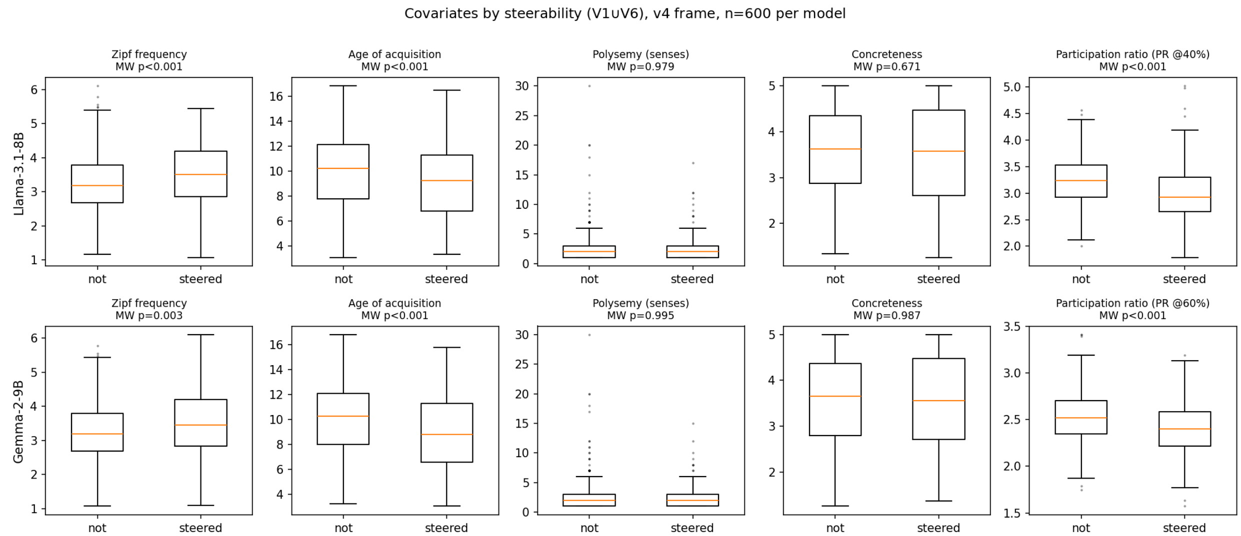

Figure 4: [**NEW jul10**] **R2/R3: covariates by steering outcome** (v4 runs, n=600 per model; PR at each model's informative depth). Zipf, AoA, and PR separate univariately in both models; polysemy and concreteness do not (their regression effects are conditional on correlated covariates).

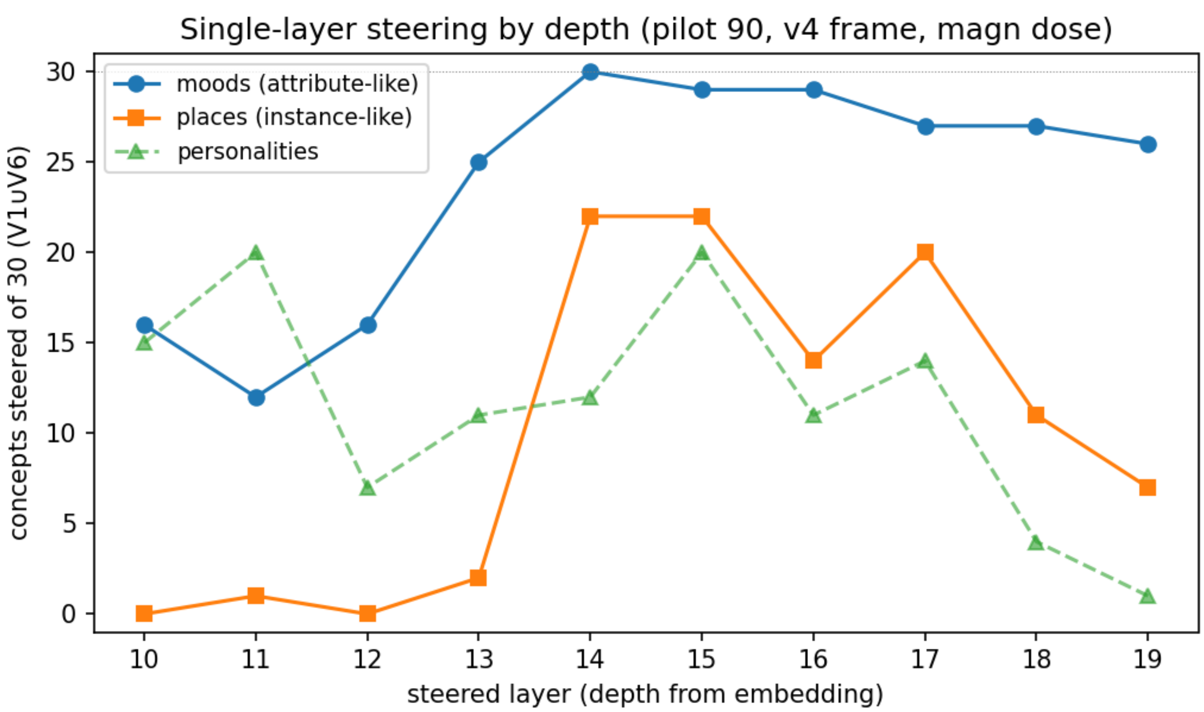

Figure 5: [**NEW jul10**] **Retention by steered depth** (pilot 90, v4 directions, single-layer magn dose). Attributes (moods) steer shallow and everywhere; entity identities (places) switch on sharply at depth 14 (of 32), window 14–17; personas never find a single-layer home.

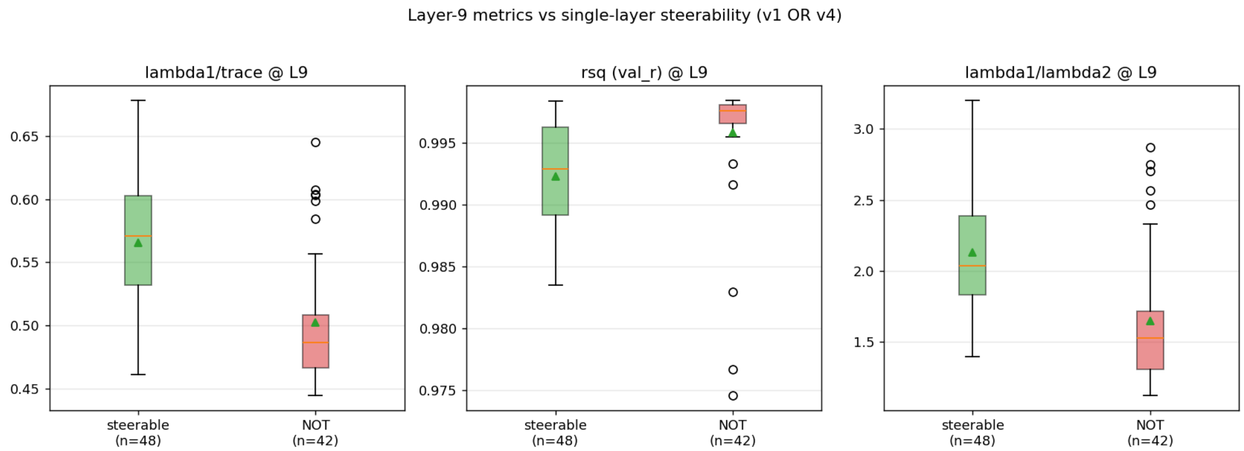

Figure 6: Appendix B: pilot spectral metrics at depth 9 by steering outcome (grouped CV-AUC ≈0.86).

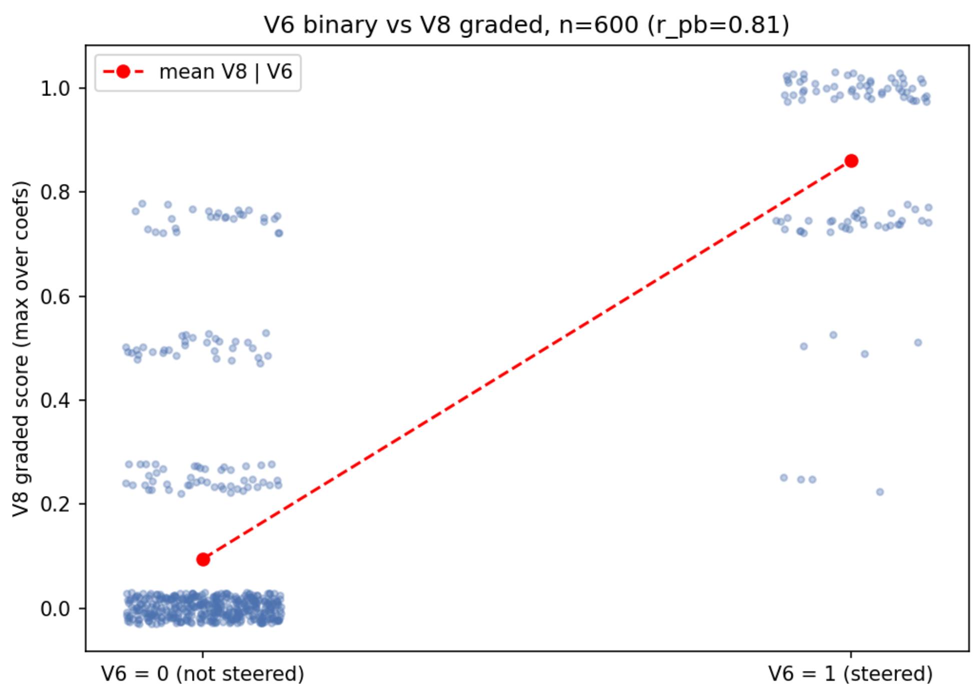

Figure 7: Appendix B: binary (V6) vs. graded (V8) judge outcomes on the same 600-run generations ($r_{pb} = 0.81$); 59 V6-failures reach V8 ≥ 0.5.

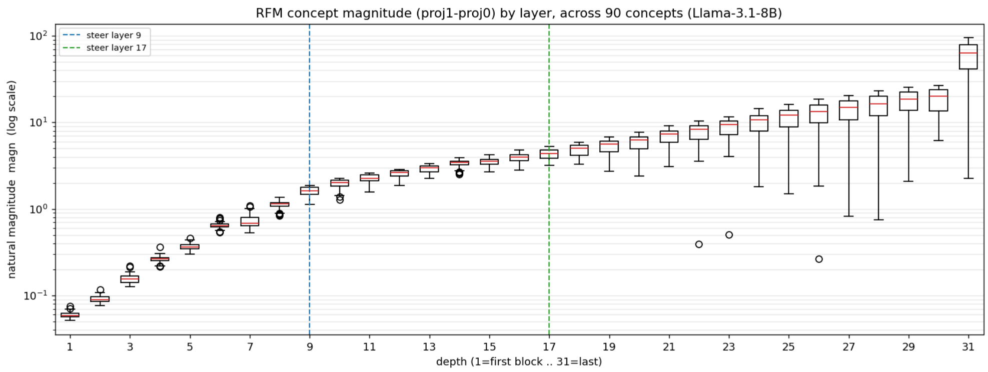

Figure 8: Appendix D: the per-layer natural projection gap (`magn`) grows ∼exponentially with depth, motivating magn-scaled dosing.

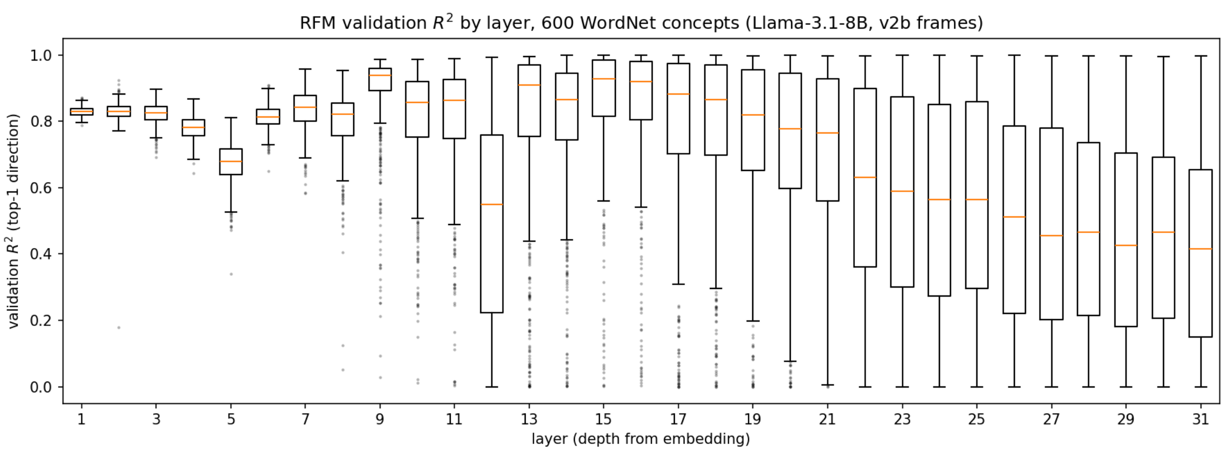

Figure 9: Appendix F: validation R^2 of the top-1 direction by layer, 600 concepts, *v2b* (dual) frames — the dual frame destroys the constant frame separator, making R^2 a per-concept signal-quality measure (wide spread, long low tail from mid-depth). Constant-prefix frames (orig, v4) saturate this metric at ∼.99 for every concept.

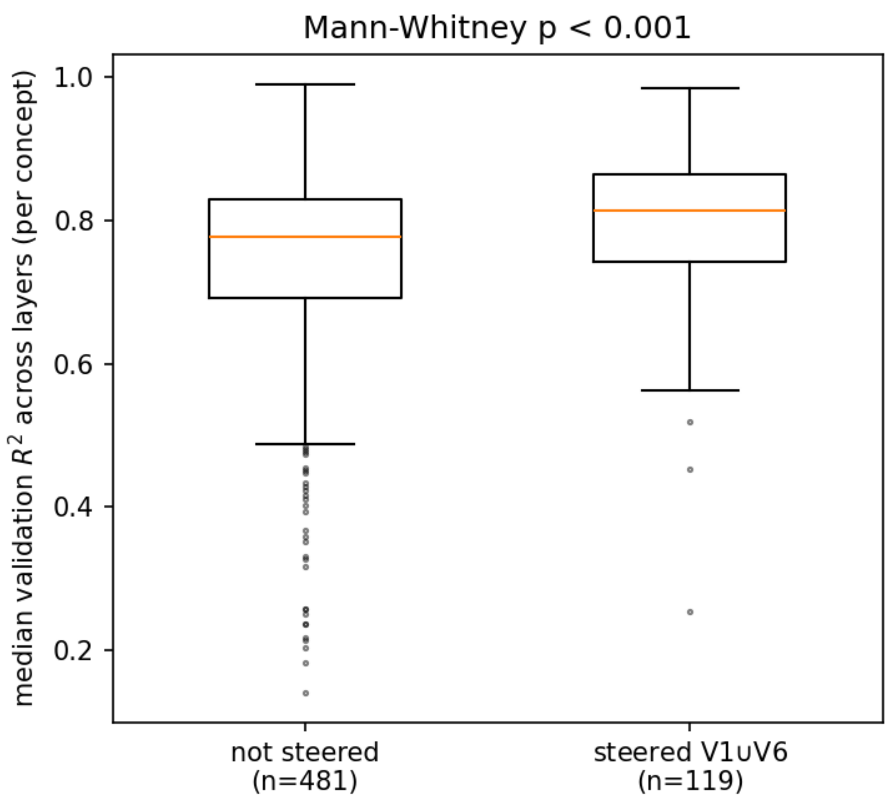

Figure 10: Appendix F: per-concept median validation R^2 (across layers, v2b frames) by steering outcome — under dual frames, extraction fit predicts steerability (MW p<.001); under constant-prefix frames the same plot is compressed to .98–1.0 and near-uninformative.